

CREDIT RISK MODELLING

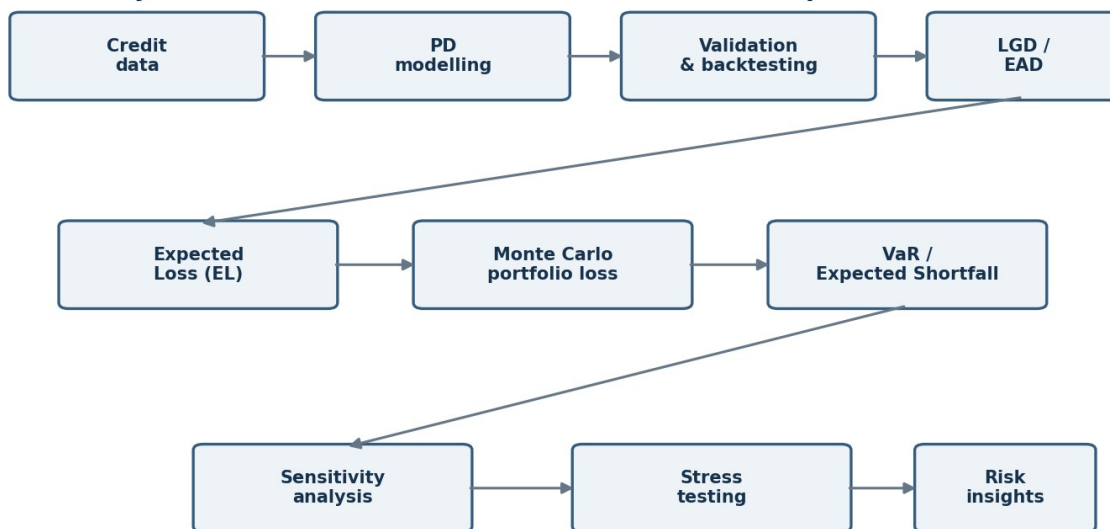
What We Did -> What We Used -> What We Got

Practical project report + interview preparation | Real LendingClub loan data

The project in one sentence

We built a borrower-level probability-of-default model, validated it on future loan vintages, converted PD into expected portfolio loss using LGD and EAD assumptions, simulated correlated defaults to study tail risk, and then performed sensitivity and stress testing.

Project workflow: borrower risk -> financial loss -> portfolio tail risk



Important scope note

This is an academic/portfolio-level risk framework using a public dataset. We should not describe it as a production banking model, regulatory validation, or a full Basel implementation. LGD and EAD are simplified because the public data do not provide the full exposure/recovery information required for production models.

1. What problem were we solving?

Credit risk has three practical layers:

- Borrower level: What is the probability that this loan defaults? -> PD
- Loss level: If it defaults, how large is the loss? -> LGD and EAD
- Portfolio level: What can happen when many loans default together? -> loss distribution, VaR and Expected Shortfall

$$PD = P(\text{Default} \mid X)$$

PD is the model-estimated probability of default for a borrower with characteristics X over the project target horizon.

$$LGD = \text{Loss} / \text{EAD}$$

LGD is the fraction of exposure that is lost after considering recoveries.

$$EAD = \text{Exposure at Default}$$

EAD is the amount expected to be outstanding when default happens. We use funded amount as a simplified proxy here.

$$EL = PD \times LGD \times EAD$$

Expected loss is the probability-weighted average financial loss. It is not a guaranteed realized loss.

Interview mental model

PD answers “Will it default?”, LGD answers “How much will we lose if it defaults?”, EAD answers “How much money is exposed?”, and the portfolio simulation answers “How bad can the combined loss become?”

2. Data we used

- Dataset: LendingClub accepted_2007_to_2018Q4.csv.gz.
- Resolved loan sample used for the project: about 1.345 million loans; about 1.077 million non-defaults and 268.6k defaults.
- Default target: Fully Paid = 0; Charged Off/Default = 1. Current/unresolved loans were excluded from this resolved-outcome setup.
- Origination features included loan amount, term, interest rate, grade/sub-grade, income, DTI, FICO, delinquencies, inquiries, revolving utilization, account counts, public records, purpose and home ownership.
- Leakage control: we did not use post-origination/post-default information such as recoveries, last payment, settlement or hardship variables inside the PD model.

Why temporal splitting matters

A random split mixes old and future behaviour. For a risk model that will be used on future loans, a time-based/out-of-time check is much more informative about model stability.

3. PD modelling: what we built

3.1 Logistic Regression

$$\text{logit}(\text{PD}_i) = \text{beta}_0 + \text{beta}_1 x_{i1} + \dots + \text{beta}_p x_{ip}$$

The model maps borrower features to default odds. $\exp(\text{beta}_j)$ is the multiplicative change in odds for a one-unit increase in feature x_j , holding other variables fixed.

- Why use it? It is interpretable and a standard baseline for credit scoring.
- What it gives us: an individual PD for every resolved loan in the validation population.
- What we learned: a simple statistical model can provide a useful risk ranking, but probability calibration must be checked separately.

3.2 LightGBM challenger

- We trained a nonlinear boosting model as a challenger model.
- Purpose: compare a transparent baseline against a more flexible machine-learning model on the same 2017 population.
- Important: a higher AUC alone does not settle model choice. Calibration, stability, interpretability, robustness and assumptions also matter.

Model	AUC	KS	Gini	Brier
Logistic	0.6999	0.2907	0.3997	0.1638
LightGBM	0.7080	0.3012	0.4159	0.1615

Practical reading

The two models are in the same performance range. LightGBM has slightly stronger discrimination on the 2017 comparison population, while the project still treats Logistic Regression as the interpretable benchmark and focuses on probability calibration and stability rather than “maximum AUC”.

4. Model validation: what did we actually check?

4.1 Discrimination

AUC = Area under the ROC curve

AUC measures ranking ability: how often a randomly chosen defaulter receives a higher risk score than a randomly chosen non-defaulter.

KS = $\max |CDF_{\text{default}} - CDF_{\text{non-default}}|$

KS measures the largest separation between cumulative score distributions of defaults and non-defaults.

Gini = $2 \times \text{AUC} - 1$

Gini is a transformed discrimination measure. It is not a separate model.

Brier = $\text{mean}((p_i - y_i)^2)$

Brier score measures probabilistic prediction error. Lower is better.

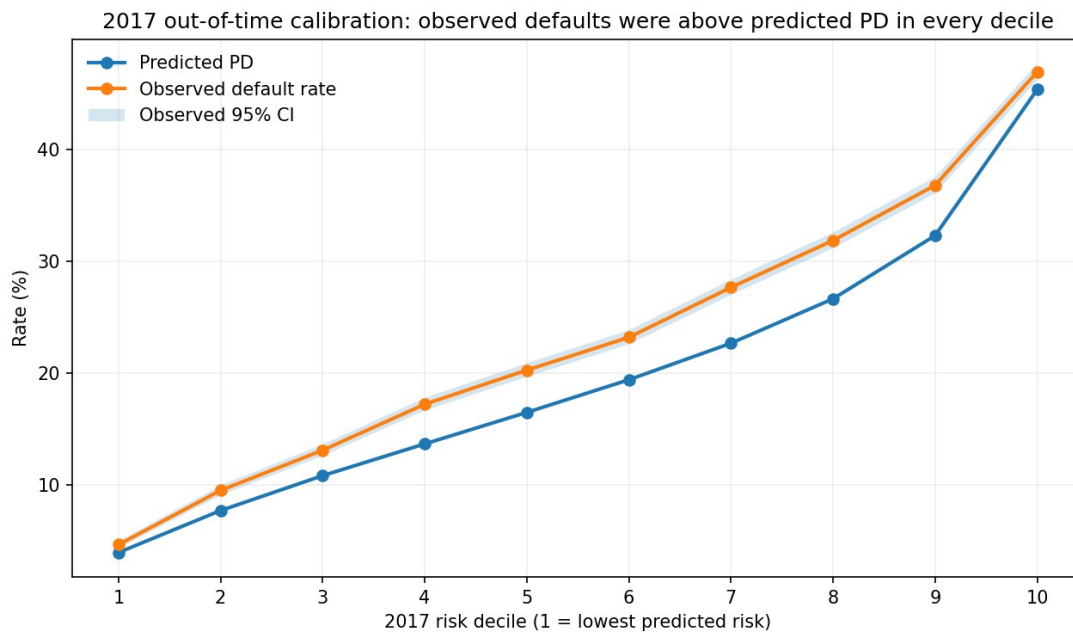
2017 extended out-of-time result

AUC = 0.6992, KS = 0.2900, Brier = 0.1639. Mean predicted PD = 19.91% versus observed default rate = 23.13%. The model therefore had useful ranking ability but underpredicted probability levels on the 2017 OOT population.

4.2 Calibration

Calibration asks: predicted PD $\approx$ observed default frequency?

If a group receives average PD 20%, we expect roughly 20% defaults over the same definition and horizon when the model is well calibrated.



What the plot tells us

Every 2017 risk decile has observed defaults above predicted PD. The gap is largest in the middle/high-risk bands, so ranking can be useful while absolute probabilities remain biased.

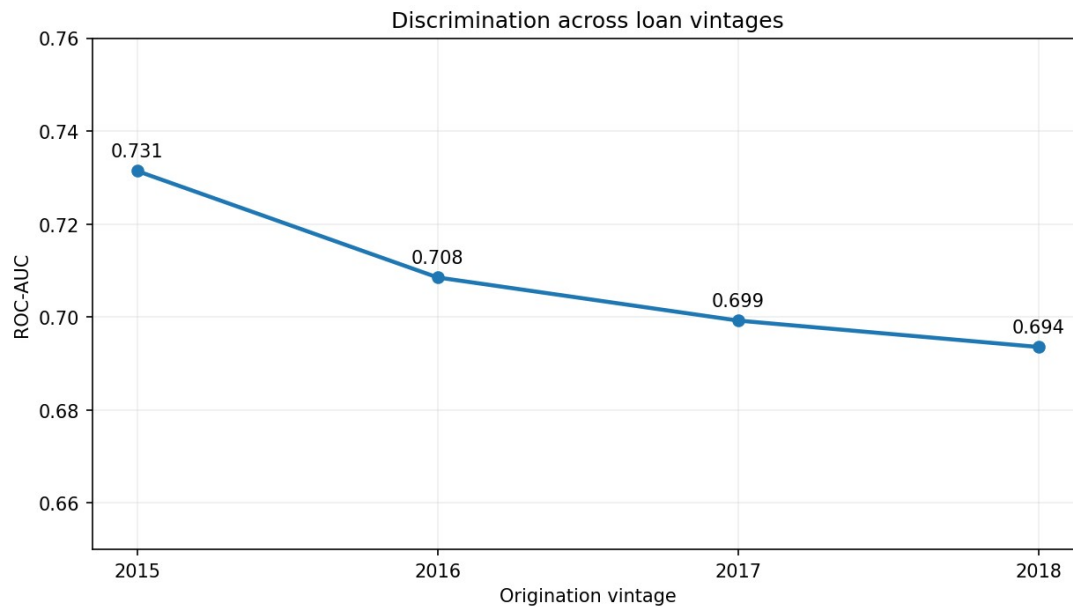
5. Backtesting and stability across vintages

5.1 What is backtesting here?

Prediction -> later actual outcome -> compare

We score historical/future vintages and compare predicted default rates with the actual outcomes that later became observable.

Vintage	Loans	Observed default	Mean PD	AUC	Brier	Actual - expected defaults
2015	375,546.0	20.19%	20.14%	0.731	0.1425	182
2016	293,105.0	23.29%	19.64%	0.708	0.1625	10,679
2017	169,321.0	23.13%	19.91%	0.699	0.1639	5,450
2018	56,318.0	15.76%	18.49%	0.694	0.1269	-1,542



2017 aggregate backtest

Expected defaults = 33,719.3; actual defaults = 39,169; gap = 5,449.7. This is an aggregate diagnostic showing the model underpredicted 2017 defaults. We treated the standardized $z \sim 34.8$ check as educational, not as a claim of full regulatory backtesting.

5.2 PSI stability

PSI = $\sum((\text{Actual}\% - \text{Expected}\%) \times \ln(\text{Actual}\% / \text{Expected}\%))$

PSI compares score-distribution shares between a reference vintage and a comparison vintage. Lower values mean less distribution shift under the chosen bins.

Vintage	PSI vs 2016
2015	0.0077
2016	0.0000
2017	0.0019
2018	0.0266

Maturity caveat

2018 has a shorter observation window, so its outcomes are not directly comparable with older vintages without maturity adjustment.

6. From PD to money: LGD, EAD and Expected Loss

6.1 LGD from realized recoveries

$$\text{LGD} = (\text{Funded Amount} - \text{Recoveries}) / \text{Funded Amount}$$

For defaulted loans, this gives the realized fraction of funded exposure that was not recovered in the available public data.

Actual project value

For 2017 defaulted loans, mean realized LGD was about 93.29%, corresponding to a recovery rate of about 6.71%.

6.2 EAD proxy

$$\text{EAD} \sim \text{Funded Amount (project proxy)}$$

The public dataset does not provide the full production exposure-at-default framework. We therefore used funded amount as a transparent proxy.

6.3 Expected Credit Loss / Expected Loss

$$\text{EL}_i = \text{PD}_i \times \text{LGD}_i \times \text{EAD}_i$$

Expected loss for loan i is the probability-weighted loss amount.

$$\text{EL}_{\text{portfolio}} = \sum_i \text{PD}_i \times \text{LGD}_i \times \text{EAD}_i$$

Portfolio expected loss is the sum of expected losses across loans.

Actual project result

Using the extended 2017 run, realized mean LGD = 93.29% and funded amount as EAD proxy gave a model-based portfolio ECL of approximately \$494.68M.

Component	Project implementation	Practical meaning
PD	Predicted by Logistic PD model	Chance of default
LGD	Realized mean from defaulted 2017 loans	Fraction of exposure lost after recoveries
EAD	Funded amount proxy	Exposure assumed at default
ECL	$\text{PD} \times \text{LGD} \times \text{EAD}$	Average probability-weighted loss

7. Why Monte Carlo? Portfolio loss and tail risk

7.1 Individual and portfolio loss

$$D_i \sim \text{Bernoulli}(\text{PD}_i)$$

$D_i = 1$ if borrower i defaults in a scenario, otherwise 0.

$$L_i = D_i \times \text{LGD}_i \times \text{EAD}_i$$

Individual scenario loss is zero if the borrower does not default and $\text{LGD} \times \text{EAD}$ if the borrower defaults.

$$L = \sum_i L_i$$

Portfolio loss is the sum of all borrower losses in one scenario.

7.2 Dependence structure

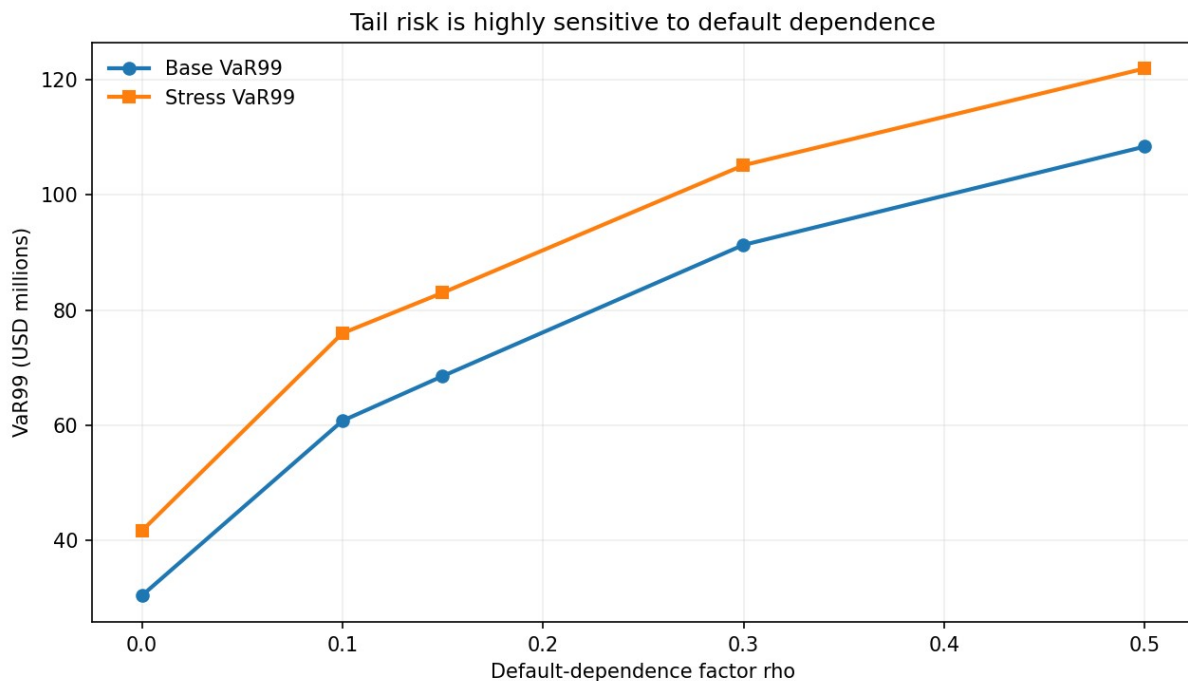
$$Z_i = \sqrt{\rho} F + \sqrt{1-\rho} \epsilonpsilon_i$$

F is a common systematic factor, $\epsilonpsilon_i$ is borrower-specific noise, and ρ controls how strongly borrower default risks move together.

- $\rho = 0$ assumes independent latent risk drivers.
- Higher ρ creates more clustered defaults in bad scenarios.
- This matters mainly for the tail of the portfolio loss distribution, not for the average loss.

7.3 Simulation setup actually used

- 10,000 actual 2017 loans sampled from the real out-of-time population for the simulation.
- 2,000 Monte Carlo scenarios.
- Base and stressed settings were evaluated at multiple dependence assumptions, including $\rho = 0.00, 0.10, 0.15, 0.30$ and 0.50 .



8. VaR and Expected Shortfall: what did we get?

VaR_alpha = alpha-quantile of the loss distribution

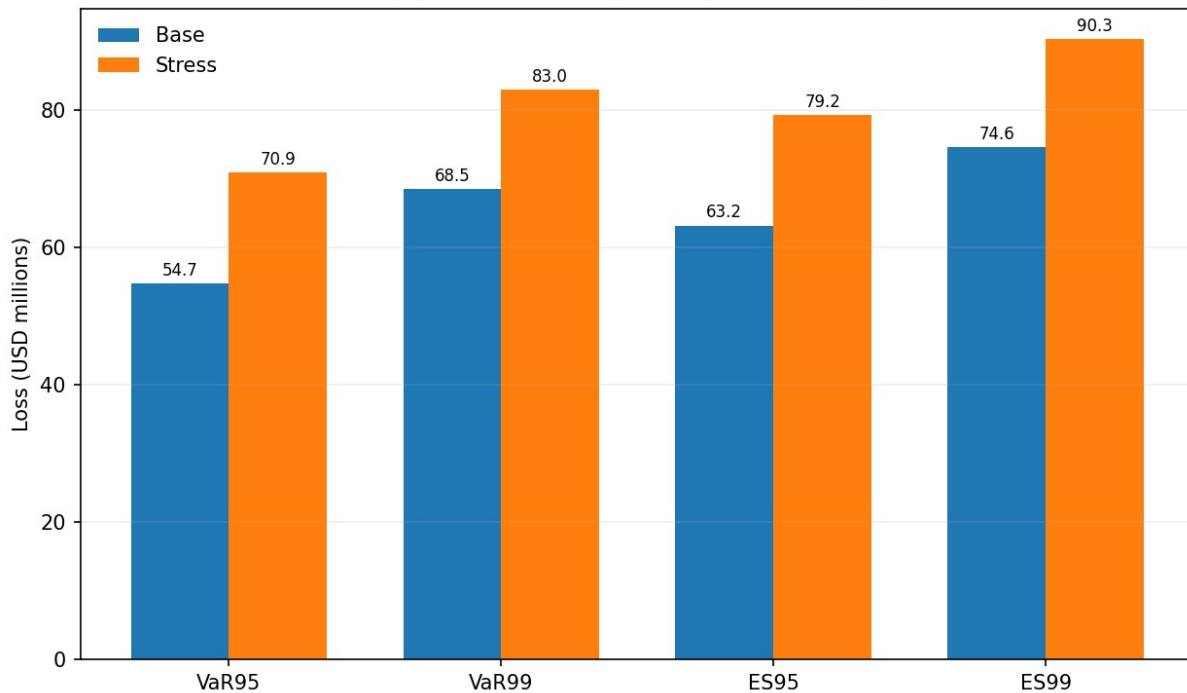
VaR is a tail threshold. VaR99 means approximately 99% of simulated losses are at or below that threshold under the model assumptions.

ES_alpha = average loss in the tail beyond VaR_alpha

Expected Shortfall tells us how severe the losses are once we enter the tail. It is generally at least as large as VaR at the same confidence level.

Scenario	Mean loss	VaR95	VaR99	ES95	ES99
Base	\$29.02M	\$54.73M	\$68.54M	\$63.18M	\$74.61M
Stress	\$40.22M	\$70.89M	\$83.01M	\$79.25M	\$90.33M

Base vs stress tail metrics at rho = 0.15



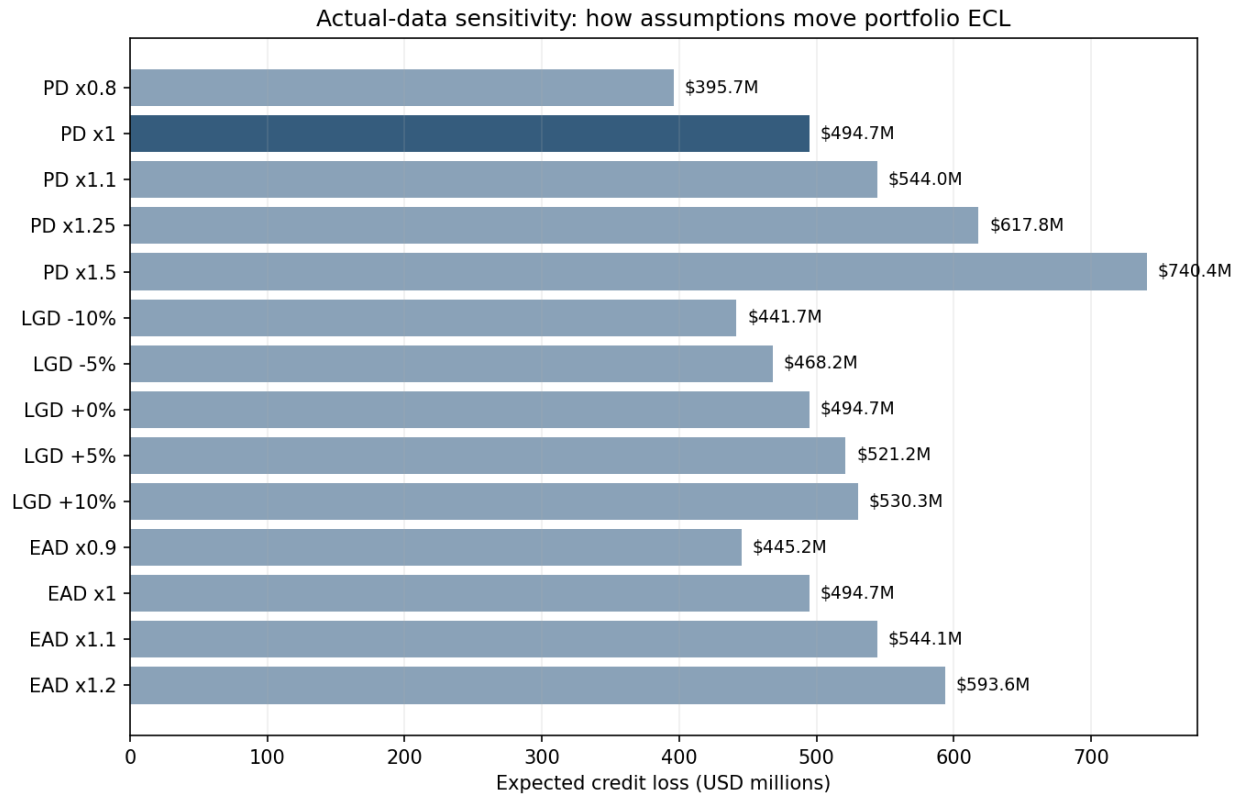
Key practical result

At rho = 0.15, the base VaR99 is about \$68.54M and ES99 is about \$74.61M. Under the stress definition used in the project, VaR99 rises to about \$83.01M and ES99 to about \$90.33M.

9. Sensitivity analysis: which assumptions move ECL?

Sensitivity = change one input/assumption -> observe output change

Sensitivity analysis is a controlled “what if” analysis, not a scenario forecast.



Change	ECL	Change vs base
PD x0.80	\$395.74M	-20.0%
PD x1.25	\$617.76M	+24.9%
PD x1.50	\$740.39M	+49.7%
LGD +5 pp	\$521.19M	+5.4%
LGD +10 pp (capped at 100%)	\$530.26M	+7.2%
EAD x1.20	\$593.62M	+20.0%

Practical interpretation

Because EL is multiplicative in PD, LGD and EAD, proportional changes in PD and EAD move ECL strongly. LGD is bounded at 100%, so large positive LGD shocks eventually have less incremental room to increase loss.

Two-factor PD-LGD stress grid

PD multiplier	LGD add	LGD used	ECL	vs base
1.00x	+0 pp	93.29%	\$494.68M	0.0%
1.00x	+5 pp	98.29%	\$521.19M	+5.4%
1.00x	+10 pp	100.00%	\$530.26M	+7.2%
1.25x	+0 pp	93.29%	\$617.76M	+24.9%
1.25x	+5 pp	98.29%	\$650.87M	+31.6%
1.25x	+10 pp	100.00%	\$662.20M	+33.9%
1.50x	+0 pp	93.29%	\$740.39M	+49.7%
1.50x	+5 pp	98.29%	\$780.07M	+57.7%
1.50x	+10 pp	100.00%	\$793.64M	+60.4%

10. Stress testing and model-risk lessons

10.1 Stress scenario used

Stress: PD odds x1.5 + LGD +5 percentage points

The project increases default odds and loss severity together to represent an adverse credit environment.

Important: odds shock is not the same as a 50% PD increase

If PD = p, odds = $p/(1-p)$. Multiplying odds by 1.5 and converting back gives a different PD than simply using 1.5p. In an interview, always distinguish PD shocks from odds shocks.

10.2 Dependence sensitivity

rho	Base VaR99	Stress VaR99
0.00	\$30.46M	\$41.69M
0.10	\$60.79M	\$75.97M
0.15	\$68.54M	\$83.01M
0.30	\$91.31M	\$105.12M
0.50	\$108.36M	\$121.90M

Main lesson

The average loss stays around \$29M in the base cases across rho values, but VaR99 rises from about \$30.46M at rho = 0 to about \$108.36M at rho = 0.50. That is a direct demonstration that dependence assumptions are a major source of model risk for portfolio tail measures.

10.3 What is not fully implemented

- No predictive LGD model with borrower-level features.
- No predictive EAD/CCF model for undrawn exposures.
- No full macroeconomic satellite model linking unemployment, rates, GDP, etc. to PD/LGD.
- No formal regulatory validation package or independent validation governance process.
- No reject-inference framework because the public dataset contains accepted loans rather than rejected applications.

11. The project results in one table

Area	What we used	What we got	What it means
PD modelling	Logistic + LightGBM	2017 AUC ~0.70	Useful but moderate risk ranking
Calibration	10 risk deciles	Observed > predicted in all deciles	Probability levels underpredicted in 2017
Backtesting	2015-2018 vintages	2017 actual - expected defaults = 5,450	OOT model underestimated 2017 defaults
Stability	Score PSI vs 2016	2017 PSI ~0.0019	Little score-distribution shift
LGD	Realized recoveries	Mean LGD ~93.29%	High realized loss severity in defaulted 2017 loans
EAD	Funded amount proxy	Base ECL ~\$494.68M	Portfolio average expected loss under assumptions
Monte Carlo	10k loans x 2k scenarios	Loss distribution	Moves beyond average loss into tail risk
VaR / ES	$\rho=0.15$	VaR99 ~\$68.54M, ES99 ~\$74.61M	Quantifies tail threshold and tail severity
Sensitivity	PD/LGD/EAD shocks	ECL up to ~\$740.39M for PD x1.5	Assumptions materially change loss
Stress	PD odds x1.5 + LGD +5pp	VaR99 ~\$83.01M at $\rho=0.15$	Adverse assumptions increase tail losses
Dependence	ρ 0 to .5	VaR99 ~\$30.46M -> \$108.36M	Correlation/default clustering drives tail risk

12. How to explain this project in an interview

45-60 second answer

"I developed a credit-risk modelling and portfolio stress-testing framework using historical LendingClub loan data. I first estimated probability of default using Logistic Regression and compared it with a LightGBM challenger. I validated the model using out-of-time AUC, KS, Gini, Brier score, calibration, vintage backtesting and PSI stability. I then combined PD with realized LGD and a simplified EAD proxy to estimate expected credit loss. To study portfolio tail risk, I simulated correlated defaults using a one-factor Gaussian dependence structure and calculated VaR and Expected Shortfall. Finally, I performed sensitivity and stress testing on PD, LGD, EAD and default dependence. The main finding was that the model ranked risk reasonably, but 2017 PDs were systematically underpredicted, and portfolio tail risk was very sensitive to the assumed default dependence."

Project logic in interview language

- "First I asked whether the model can rank risky borrowers." -> AUC / KS / Gini.
- "Then I asked whether the probabilities are numerically right." -> calibration.
- "Then I asked whether the model works on future vintages." -> OOT backtesting.
- "Then I asked whether the model is stable." -> vintage analysis / PSI.
- "Then I converted default probability into money." -> PD x LGD x EAD.
- "Then I asked what a bad portfolio can look like." -> Monte Carlo -> VaR / ES.
- "Finally I challenged assumptions." -> sensitivity + stress + dependence analysis.

13. Interview questions an interviewer can ask

A. Basic project questions

Q. What is the objective of your project?

A. To connect borrower-level default probability modelling with portfolio-level expected loss and tail-risk analysis.

Q. Why did you choose LendingClub data?

A. It provides a large historical loan-level dataset with borrower, loan and outcome information, which lets us build a complete educational credit-risk workflow on real observations.

Q. What is your target variable?

A. A binary default outcome: Fully Paid = 0 and Charged Off/Default = 1. Unresolved Current loans are excluded from this resolved-outcome setup.

Q. Why did you exclude current loans?

A. Their final outcome is not yet known in the resolved dataset setup, so using them as definite non-defaults would contaminate the target with unresolved observations.

Q. Why did you remove post-default variables?

A. Because they would leak information from after the prediction point. A model must use only information available at the time the prediction is supposed to be made.

Q. Why Logistic Regression first?

A. It is interpretable, statistically familiar and a common baseline for default modelling. It gives coefficient-level reasoning and a useful benchmark for a more complex challenger.

Q. Why compare LightGBM?

A. To test whether nonlinear interactions can improve ranking relative to the interpretable baseline.

Q. What did you get from the comparison?

A. On the same 2017 population, Logistic AUC was 0.6999 and LightGBM AUC was 0.7080; the corresponding KS values were 0.2907 and 0.3012.

B. Model validation questions

Q. What is AUC?

A. AUC measures discrimination or ranking ability. It can be interpreted as the probability that a randomly selected defaulter receives a higher risk score than a randomly selected non-defaulter.

Q. What is KS?

A. KS is the maximum vertical separation between the cumulative score distributions of defaults and non-defaults.

Q. Why is AUC not enough?

A. AUC tells us whether high-risk borrowers are ranked above low-risk borrowers. It does not tell us whether a predicted PD of 20% actually corresponds to about 20% observed defaults.

Q. What is calibration?

A. Calibration checks whether predicted PD values agree with observed default frequencies for appropriately defined groups and horizons.

Q. What did calibration show?

A. On 2017 OOT data, every one of the ten risk deciles had an observed default rate above the average predicted PD.

Q. Mean predicted PD was 19.91% and observed default was 23.13%. What does that mean?

A. The model systematically underestimated the absolute probability of default on the 2017 OOT population.

Q. What is the calibration slope/intercept telling you?

A. The extended 2017 calibration intercept was about 0.078 and slope about 0.896. Together they indicate a level shift and some deviation in calibration around the observed outcomes; the practical finding is the systematic underprediction seen in the deciles.

Q. What is backtesting?

A. It is comparing model predictions with subsequently observed outcomes.

Q. Why use out-of-time validation instead of only a random train-test split?

A. Because time-based validation tests whether a model trained on past behaviour still works on future behaviour and can reveal temporal drift.

Q. What did the 2017 backtest show?

A. Expected defaults were about 33,719 while actual defaults were 39,169, a gap of about 5,450 defaults.

C. Credit-risk formula questions

Q. What is PD?

A. PD is the probability that a borrower defaults over a specified horizon, conditional on the available borrower information.

Q. What is LGD?

A. Loss Given Default is the fraction of exposure lost after recoveries when default happens.

Q. What is EAD?

A. Exposure at Default is the amount expected to be outstanding at the time of default.

Q. Give the expected loss formula.

A. $EL = PD \times LGD \times EAD$.

Q. Derive EL from probability.

A. If loss is $LGD \times EAD$ when default occurs and zero otherwise, then $E[L] = P(\text{Default})(LGD \times EAD) = PD \times LGD \times EAD$.

Q. What LGD did your project use?

A. The mean realized LGD among defaulted 2017 loans was about 93.29%, computed from funded amount and recoveries.

Q. How did you model EAD?

A. We used funded amount as a simplified EAD proxy because the public dataset does not give the complete production exposure framework.

Q. Why is PD alone insufficient?

A. Two borrowers can have the same PD but very different LGD and EAD, so their expected financial losses can be very different.

Q. What was your portfolio ECL?

A. Approximately \$494.68M in the extended 2017 run under the project assumptions.

Q. Is expected loss the actual loss?

A. No. It is an average probability-weighted loss. Realized loss in one future period can be higher or lower.

D. Monte Carlo / VaR / ES questions

Q. Why Monte Carlo?

A. Expected loss gives only the average. Monte Carlo lets us generate many possible default scenarios and study the full portfolio loss distribution.

Q. How do you simulate a default?

A. For a simple independent model, draw $U \sim \text{Uniform}(0,1)$ and set $D_i = 1$ when $U < PD_i$. In our extended portfolio simulation we used a one-factor Gaussian dependence construction before converting latent scores into default events.

Q. Why model dependence between borrowers?

A. Real borrowers can be exposed to common economic conditions, so defaults may cluster. Ignoring that dependence can materially underestimate tail losses.

Q. What is the one-factor structure?

A. $Z_i = \sqrt{\rho}F + \sqrt{1-\rho}\epsilon_i$, where F is common systematic risk and ϵ_i is borrower-specific risk.

Q. What is VaR99?

A. It is the 99th percentile of the simulated loss distribution. Roughly 99% of simulated losses are at or below this threshold, under the stated assumptions.

Q. Is VaR the maximum loss?

A. No. VaR is a quantile threshold, not the maximum possible loss.

Q. What is Expected Shortfall?

A. It is the average loss in the tail beyond the VaR threshold.

Q. What were your base VaR99 and ES99 at $\rho = 0.15$?

A. \$68.54M and \$74.61M, respectively, in the extended run.

Q. What happened under stress at $\rho = 0.15$?

A. VaR99 increased to about \$83.01M and ES99 to about \$90.33M.

E. Attack / cross questions

Q. Your AUC is only about 0.70. Why is that acceptable?

A. I would not label it as “good” in isolation. I would say it provides moderate discrimination on the chosen population, and the project therefore relies on a broader validation framework rather than AUC alone.

Q. LightGBM has higher AUC. Why not use LightGBM automatically?

A. Because credit model selection is multi-dimensional. We also care about calibration, stability, interpretability, assumptions, robustness and governance.

Q. Your model is underpredicting 2017 defaults. What would you do next?

A. Investigate calibration by vintage and segment, check whether the development population differs from 2017, assess recalibration options such as intercept/slope adjustment, and test the corrected model on another out-of-time period.

Q. Why can a model have good AUC but poor calibration?

A. Ranking and probability scaling are different properties. A monotonic transformation can preserve ordering while changing probability values.

Q. Why is 2018 tricky?

A. Later-vintage loans have shorter maturity and outcome observation windows, so they are not fully comparable with older vintages for default outcome assessment.

Q. Is the $z \sim 34.8$ a regulatory backtest?

A. No. It is a simple educational aggregate diagnostic. A formal validation package would require more careful treatment of horizon, dependence, calibration, confidence intervals and governance.

Q. What is the biggest modelling limitation?

A. LGD and EAD are simplified rather than independently predictive models, and the portfolio dependence structure is a modelling assumption rather than an observed structural estimate.

Q. What would you build next?

A. A borrower-level LGD model, a predictive EAD/CCF model, maturity-aware outcome definitions, richer calibration testing and macro-linked scenario generation.

Q. What is reject inference and why can't you do it here?

A. Reject inference tries to account for applicants who were rejected and therefore do not have observed loan outcomes. The public accepted-loan dataset does not give a reliable observed outcome for rejected applications.

Q. What is the difference between sensitivity analysis and stress testing?

A. Sensitivity changes one input or assumption and observes the output response. Stress testing applies a predefined adverse but plausible scenario, often changing several linked assumptions together.

F. Very common “trap” questions

Q. If PD = 10%, what are the odds?

A. Odds = $0.10 / 0.90 = 0.1111$.

Q. If odds = 1, what is PD?

A. PD = odds / (1 + odds) = $1/2 = 50\%$.

Q. If PD is multiplied by 1.5, is that the same as multiplying odds by 1.5?

A. No. They are different transformations. Odds = $p/(1-p)$, so an odds shock must be converted back to PD.

Q. If AUC = 0.75, what is Gini?

A. Gini = $2 \times 0.75 - 1 = 0.50$.

Q. If PD = 5%, LGD = 40%, EAD = 10 lakh, what is EL?

A. $0.05 \times 0.40 \times 10,00,000 = \text{Rs } 20,000$.

Q. If VaR99 = \$50M, what does it mean?

A. Approximately 99% of losses are at or below \$50M and approximately 1% exceed it, under the stated model, horizon and assumptions.

Q. If VaR99 = \$50M and ES99 = \$70M, which is larger and why?

A. ES99 is larger because it averages the losses deeper in the 1% tail, while VaR99 is only the threshold where that tail begins.

Q. Why does higher rho increase VaR much more than mean loss?

A. The mean depends mainly on marginal PD, LGD and EAD. Correlation changes how often losses cluster in the same scenario, which strongly changes tail concentration.

Q. Can two portfolios have the same expected loss but different VaR?

A. Yes. They can have the same average loss but different concentration, dependence and tail behaviour.

Q. What is model risk?

A. The risk that a model's assumptions, specification, implementation or use do not adequately represent the real-world process and therefore lead to misleading decisions.

14. What the interviewer should remember about this project

Core message

This is not “just a classification project”. The interesting part is the complete quantitative-risk chain: default prediction -> model validation -> probability calibration -> expected credit loss -> portfolio loss simulation -> tail measures -> sensitivity -> stress testing.

- Strongest evidence from actual data: 2017 OOT calibration underprediction and multi-vintage backtesting.
- Strongest quantitative finance element: PD x LGD x EAD, Monte Carlo loss distribution, VaR and Expected Shortfall.
- Strongest model-validation element: discrimination + calibration + backtesting + benchmarking + stability + assumption sensitivity.
- Best interview habit: always state the population, horizon/setup and assumptions before quoting a number.
- Do not overclaim production/regulatory status. Treat simplified LGD/EAD and dependence as explicit project assumptions.

One-line closing answer

“The project moved from who is likely to default to how much the portfolio could lose, including under stressed and correlated default scenarios.